

Cite this: DOI: 00.0000/xxxxxxxxxx

Unraveling the role of chemical composition in the lattice thermal conductivity of oxychalcogenides as thermoelectric materials assisted by machine learning.[†]Ernesto J. Blancas,^a Jose J. Plata,^{*a} Julia Santana,^a Felipe Lemus Prieto,^b Antonio M. Márquez,^a and Javier Fdez. Sanz^aReceived Date
Accepted Date

DOI: 00.0000/xxxxxxxxxx

Oxichalcogenides represent a large chemical space with potential application as thermoelectric materials due to their low thermal conductivity. However, the nature of this behaviour is still under debate. Understanding the origin of the anharmonicity of these materials is key to developing models that improves the efficiency of thermoelectric materials. In this work, we combine machine learning with first principles calculations to explore oxichalcogenides materials. Machine learning not only accelerates the prediction of the lattice thermal conductivity for large chemical spaces with high accuracy, but also catalyzes the development of design principles to discover new thermoelectric materials. Using this approach, lattice thermal conductivity has been directly connected to the effect of each species in the material, using atomic projections of the scattering rates. The role of the monovalent atom and the lone pair electron for the trivalent cation are discussed in detail. Based on this knowledge, it is possible to connect complex properties such as lattice thermal conductivity with a more manageable synthetic variable such as chemical composition. Using this strategy, we propose promising new oxichalcogenides such as BioAgSe, which subsequently has been confirmed as having ultra-low lattice thermal conductivity.

1 Introduction

Global energy demands require the development of new strategies to improve the efficiency of our energy grid and, simultaneously, accelerate the transition to renewable energies. Thermoelectric, TE, materials are one of the best available solutions when it comes to reducing the dependency on fossil fuels and making other green energy resources more cost efficient. However, the discovery and optimization of new thermoelectric materials face many obstacles. The efficiency of a thermoelectric materials, measured by its figure of merit ZT , rarely surpassed 1 during the last 50 years of the 20th century. This is partially due to the relationship between the transport properties that define ZT ,

$$ZT = \frac{S^2 \sigma T}{\kappa_e + \kappa_l}, \quad (1)$$

where σ is the electrical conductivity, κ_e and κ_l are the electronic and lattice thermal conductivities, T is the temperature and S is

the Seebeck coefficient. Maximizing ZT is a difficult and complex task; Wiedemann-Franz law requires σ to be proportional to κ_e and Pisarenko relation limits the enlargement of S and σ simultaneously.^{1,2} For this reason, one of the most common strategies for maximizing ZT is looking for new thermoelectric materials with low κ_l .

The experimental measurement of κ_l or any other of the transport properties involved in ZT represent another obstacle. Most of these properties are extremely sensitive to synthesis conditions, hampering the systematic exploration of large chemical spaces. Despite of these difficulties, thermoelectric materials are experiencing a renaissance during last two decades.³ Atomistic simulations and computational materials science have propelled the discovery of many TE materials with ZT values above 2.⁴ Nevertheless, there is plenty of room for improvement in this area. For instance, the accurate prediction of κ_l requires large and expensive ab-initio molecular dynamics, MD,⁵ or the calculation, at least, of 3rd-order interatomic forces constants, IFCs.⁶

Machine learning, ML, has recently emerged as the ideal strategy for reducing computational costs when calculating κ_l . Accurate ML-based potentials have reduced the computational cost of approaches based on MD⁷ and new methods have been developed to extract nth-order IFCs using ML at a reduced cost.^{8,9} These new approaches have changed the paradigm when exploring new TE

^a Departamento de Química Física, Universidad de Sevilla, Seville, Spain; E-mail: jplata@us.es

^b Research, Technological Innovation and Supercomputing Center of Extremadura (Cen-itS), Cáceres, Spain

[†] Electronic Supplementary Information (ESI) available: [details of any supplementary information available should be included here]. See DOI: 00.0000/00000000.

materials. Exploring large chemical spaces is no longer a problem, and the scientific community is focusing on extracting design principles for these large amount of data.¹⁰ However, this last step cannot be underestimated. Connecting real space variables such as chemical composition with reciprocal-space variables which determine transport properties is fundamental for developing new and more efficient TE materials.¹¹

Oxychalcogenides represent one of these large chemical spaces that need to be explored in more detail.¹² Since BiOCuSe was reported as a promising TE material in 2011,^{13,14} a large amount of data has been published with respect to their TE performance.^{15–17} For instance, different strategies have been proposed to increase their power factor, such as doping,¹⁸ alloying¹⁹ or engineering grain boundaries.²⁰ Torimayama *et al.* have recently demonstrated that electron transport properties of these materials can be improved through n-type doping instead of p-type doping.²¹ However, the key property that make oxychalcogenides excel as TE materials is their low lattice thermal conductivity. Interestingly, this fact has been attributed to different factors, and there is still some controversy. For instance, Ding *et al.* described how anharmonic vibrations and structural scattering of phonons are partially caused by in-layer and interlayer off-phase vibrational modes.²² Saha *et al.* compared the vibrational properties of BiOCuSe and LaOCuSe, finding that the main reason for their different κ_l is the mass difference between Bi and La.^{23,24} Additionally, Vaqueiro *et al.* combined DFT calculations with analysis of in-situ neutron diffraction data to demonstrate that weak bonding of copper atoms play a crucial role, reducing the lattice thermal conductivity of these compounds.²⁵ Although all these works present solid proofs, most of them are based on the study of harmonic (phonon dispersion curves) or quasiharmonic (Grüneisen parameters) properties. Lattice thermal conductivity is a property that stems from the anharmonicity of the material, thus, features such as scattering rates are essential to understanding its behavior. However, until recently, exploring highly computationally demanding properties such a κ_l in large chemical spaces was an unaffordable task.

Here, we present a new approach in which the low thermal conductivity of oxychalcogenides is directly connected to their chemical composition. Machine Learning is combined with first principles calculations to accelerate the calculation of κ_l for oxychalcogenides. These results are analyzed through atomic-projected scattering rates-frequency heat maps, unraveling the role of each element in the anharmonicity and lattice thermal conductivity of these materials.

2 Methodology

Geometry optimization. The atoms and the lattice of the ground state structures were fully relaxed at 0 K with VASP package,^{26,27} using projector-augmented wave (PAW) potentials.²⁸ The exchange-correlation functional proposed by Perdew-Burke-Ernzerhof (PBE)²⁹ was combined with Grimme D3 Van der Waals corrections³⁰ to obtain the energies. D3 corrections are critical for a good description of slab interactions and lattice parameters (Table 1). Core and valence electrons were selected following standards proposed by Calderon *et al.*³¹ Plane waves with a cut-off of 500 eV were used for valence electrons and the primitive cells

were optimized with a $5 \times 5 \times 3$ k-point mesh. Wavefunction was considered as converged when the energy difference between two electronic steps was smaller than 10^{-9} eV. Geometry and lattice vectors were fully relaxed, using an 8 atoms conventional cell, until forces over all atoms were smaller than 10^{-7} eV/Å. An additional support grid for the evaluation of the augmentation charges was included to reduce the noise in the forces.

Table 1 Calculated lattice parameters a and c for all the materials explored in this work. Average lattice thermal conductivity, κ_l , and its components in xx and zz directions at 300 K. Experimental values are included in parenthesis.

Material	a (Å)	c (Å)	κ_l (W K ⁻¹ m ⁻¹)	κ_l^{xx}	κ_l^{zz}
BiOCuSe	3.91 (3.92) ³²	8.97 (8.91) ³²	1.26 (1.03) ¹⁴	1.66	0.47
BiOCuS	3.85 (3.87) ³²	8.59 (8.56) ³²	1.83	2.20	1.08
LaOCuS	3.94 (4.00) ³³	8.45 (8.51) ³³	4.41	5.40	1.64
AcOCuS	4.045	8.64	1.70	2.44	0.24
LaOAgS	4.03 (4.05) ³⁴	9.04 (9.04) ³⁴	2.45	3.28	0.80
LaOLiS	3.98	8.70	7.83	9.76	3.97
BiOAgSe	3.99 (3.99) ³⁵	9.54 (9.46) ³⁵	0.87	1.09	

Supercell single-point calculations and force constants. Inter-atomic force constants, IFCs, were calculated using the HiPhive package, which combines the forces calculated for random atomic distortion in supercells with machine learning regression.⁸ The forces were calculated in $5 \times 5 \times 3$ supercell (600 atoms) using the same setup as the one used for the geometry optimizations. The amplitude of the distortions applied to the atoms plays an important role in the calculation of the IFCS so a 2 steps approach was designed.³⁶ First, small random distortions were generated for all the atoms of 3 supercells and second and third-order IFCs were calculated using the HiPhive package. Then, eight new distorted supercells were created superimposing normal modes with random phase factors and amplitudes corresponding to 300 K. The recursive feature elimination RFE, algorithm was used to obtain the force constants from the DFT forces. Even though RFE is more expensive than ordinary least-square regression, this algorithm requires fewer number of structures to converge. Additionally, reducing the number of parameters via RFE also simplifies the model, keeping only the most relevant interaction terms. IFCs were calculated including cutoffs for second, third and fourth-order terms. In order to ensure transferability across compounds, cutoffs were determined based on coordination shells. We have used a wrapper code for the hiPhive program that automates the distorted supercell creation, force calculation using VASP, and the construction of the ML IFCSs, which is available for public use.^{36,37}

BTE solver. The ShengBTE package was used to solve the BTE and obtain lattice thermal conductivity.³⁸ BTE is solved through the iterative approach, which predicts more accurate values than the relaxation time approximation. Average lattice thermal conductivity

ity, κ_l , will be defined as the average of the diagonal values of the $\kappa_l^{\alpha\beta}$ tensor. Total scattering rates, Γ_{tot} , were computed including isotopic and three-phonon scattering. Memory demand and the convergence of κ_l with the number of $\mathbf{q}$ -points were balanced using a gaussian smearing of 0.1 eV and a dense mesh of $12 \times 12 \times 12$. Scattering rates heat maps and their projections were obtained combining ShengBTE and Phonopy³⁹ through a high-throughput tool which is available for public use.⁴⁰

3 Results

BiOCuSe. BiOCuSe is a layered oxychalcogenide that belongs to the tetragonal structural prototype ZrCuSiAs. Its 3D structure alternates fluorite-like $[\text{Bi}_2\text{O}_2]^{2+}$ slabs with antiferro-like $[\text{Cu}_2\text{Se}_2]^{2-}$ slabs along the c -axis.¹² In order to give a good description of the long range interactions between layers, Grimme D3 Van der Waals corrections³⁰ have been included, obtaining very good agreement with both, a and c , experimental lattice parameters for BiOCuSe and the other oxychalcogenides included in this work (Table 1). Dispersion curves extracted from ML-regression are also in good agreement with the results obtained using the finite differences approach, full-DFT, where the IFCs are determined directly from the displacement of one or two specific atoms in the supercell (Fig. 1a). Root mean square error, RMSE, of

the forces rapidly converges with respect to the number of supercells included in the training (Fig. S1a). Average lattice thermal conductivity, κ_l , follows the same trend and it is aligned with previous values calculated using more computationally-demanding approaches.⁶ Cutoff-interaction distances of 9.75, 6.89, and 3.71 Å for the second-, third-, and fourth-order force constants were found to be sufficient for convergence of the ML-based method. These cutoffs were consistently modified for other materials based on coordination shells. Lattice thermal conductivity was also converged with respect to the $\mathbf{q}$ -points grid size (Fig. S1b). Using a ML-model to extract IFCs and compute κ_l drastically reduces the computational cost without reducing the accuracy of the predictions. Small differences with previous calculated κ_l for BiOCuSe are due to the use of different exchange correlation functionals.⁴¹ Kumar *et al.* obtained slightly larger lattice parameters using a PBE+U functional which usually is connected to less rigid bonds and lower κ_l . Both, ML-model and full-DFT results, slightly overestimate experimental values,¹⁴ which can be justified because of the polycrystallinity of the samples and the presence of defects that can reduce κ_l (Fig. 1b).

Effects of chalcogen substitution. It has been already reported that κ_l follows a monotonous trend down the chalcogen group for BiOCQ (Q = S, Se, Te).^{17,42} This is not always the case. For instance, I-III-VI₂ chalcopyrite semiconductors follows a non-monotonous variation of κ_l , with selenides being the materials with the lowest values.³⁶ Substituting Se by S brings strong modifications in the phonon density of states, DOS (Fig. 2a-b). While vibrational modes of the $[\text{Cu}_2\text{S}_2]^{2-}$ slab are strongly localized on Cu at lower frequencies (around 2 THz) or S (around 6-8 THz), $[\text{Cu}_2\text{Se}_2]^{2-}$ slab presents more delocalized vibrations between Cu and Se, participating both atoms in vibrational modes at 2 and 4-5 THz. There are not significant differences between the group velocities (Fig. S2) of BiOCuSe and BiOCuS so their differences in κ_l should be connected to the anharmonicity of the systems. Although they can look similar, there are important differences in the scattering rates of both compounds in the 2-5 THz region of the spectra, where acoustic-optical phonon scattering takes place (Fig. 2c). Increasing the atomic mass of the chalcogen delocalizes and reduces the frequencies of the vibrational modes at which the $[\text{Cu}_2\text{Q}_2]^{2-}$ slabs are vibrating. This shift maximized the overlap of these optical vibrational modes with the acoustic modes that are mainly localized at the Bi atoms, increasing their scattering rates and reducing κ_l . The same behavior can be extracted from the average accumulative κ_l where there are not big differences in κ_l between BiOCuSe and BiOCuS up to 2 THz approximately (acoustic region) (Fig. 2d). The lattice thermal conductivity of both compounds changes in the region where acoustic-optic scattering takes places.

Effects of the trivalent cation. While there is consensus about the role of the chalcogen atom and how its chemical substitution can slightly modify κ_l for BiOCuQ (Q = S, Se, Te) compounds, the role of the Bi is certainly under debate. Some authors point to the Bi atom's lone pair electron as the main reason for the low thermal conductivity of BiOCuSe,^{15,43} but there are also studies that suggest that the Cu-Q vibrations are the ones which contribute to reduce κ_l .²⁵ Partially, these discrepancies come from

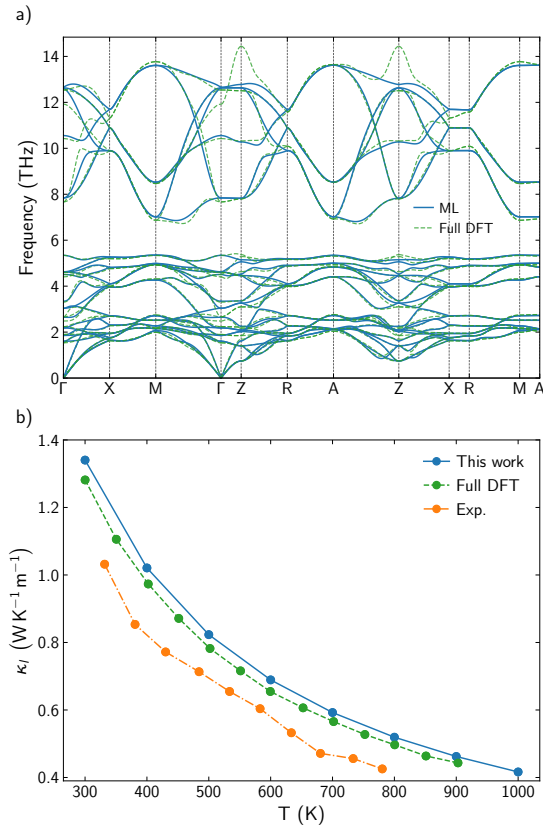


Fig. 1 (a) Comparison of BiOCuSe phonon dispersion curves obtained using the machine-learned force constant potential, ML, via HiPhive with those from the full-DFT method via Phonopy. (b) Comparison of experimental κ_l (orange)¹⁴ with calculated results using a full-DFT approach (green),⁴¹ and ML-based model (blue) between 300 and 900 K.

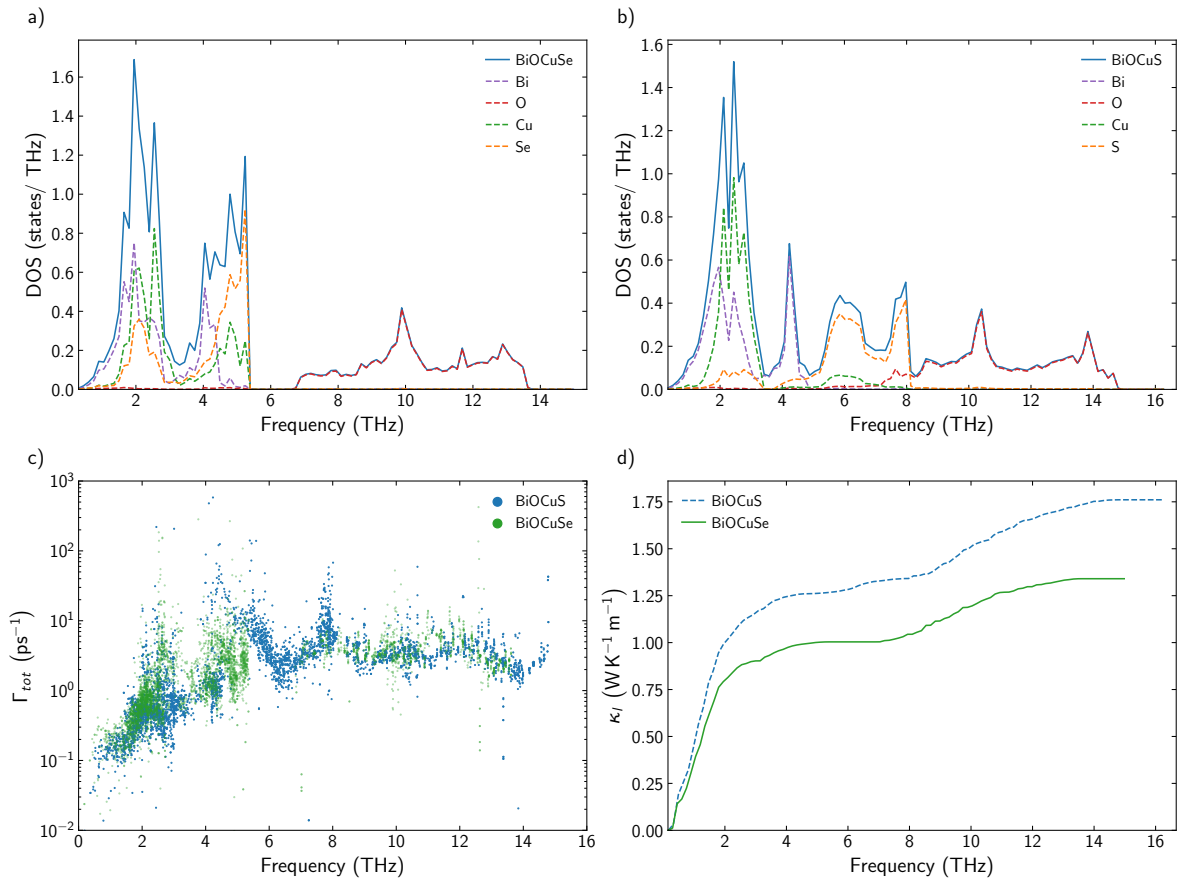


Fig. 2 (a-b) Total and projected phonon density of states, DOS, of BiOCuSe and BiOCuS. (c) Total scattering rates, Γ_{tot} for BiOCuSe (green) and BiOCuS (blue) at 300 K. (d) Cumulative κ_l for BiOCuSe (green solid line) and BiOCuS (blue dashed line) at 300 K.

using only harmonic (phonon dispersion curves) or quasiharmonic (Grüneisen parameters) features to explore a property that is strongly coupled to the anharmonicity of the system. Analyzing the scattering rates is fundamental to understanding the origin of the low lattice thermal conductivity of oxychalcogenides. In order to understand the role of Bi, previous works have compared the vibrational, elastic and electronic properties of the BiOCuQ materials with their La counterparts.^{23,24} However, that substitution introduces more than one change in the system: i) the lone pair is removed and ii) the mass of the trivalent atom is drastically reduced. Here, the lattice thermal conductivity and anharmonicity of BiOCuS, LaOCuS and AcOCuS are compared in order to decouple the role of the mass and lone pair of the trivalent atom. In addition to the phonon DOS and the calculation of the cumulative κ_l (Fig. 3), scattering rates - frequencies heat maps have been built, including its projection over each element (Fig. 4). These representations not only connect phonon frequencies with a direct measurement of the anharmonicity of the system, but they also unravel the role of each species. Phonon density of states, DOS, of LaOCuS and AcOCuS present similarities with BiOCuSe DOS (Fig. 3a-b). All of them present a region around 2 THz where optics bands are overlapping with acoustic bands, participating Cu and the trivalent cation. However, DOS cannot explain the differences in the in-plane (xx) and perpendicular (zz) cumulative κ_l . Due to

the layered nature of these materials, κ_l^{zz} which is perpendicular to the slab stacking, is smaller than κ_l^{xx} (Fig. 3c-d). Cumulative κ_l^{zz} follows the same trend for all three materials, 100% of κ_l^{zz} is stored up to 2 THz, approximately. The acoustic-optic scattering region around 2 THz includes atoms from both different layers which can be related to the interlayer off-phase vibrational modes described by Ding.²² Larger κ_l^{zz} values are obtained for LaOCuS, then BiOCuS and finally AcOCuS which points to the atomic mass as the main responsible of this trend. However, the behavior is more complex for κ_l^{xx} . Cumulative κ_l^{xx} presents two main contributions, one in the 0-3 THz range and another one around 8-13 THz (Fig. 3c-d). First large contribution to κ_l^{xx} follows the same trend than κ_l^{zz} , ranking their values based on the atomic mass of the trivalent atom. In addition to larger group velocities, heat maps present lower scattering rates in the 0-3 THz range for LaOCuS, which is linked to a lower anharmonicity and larger κ_l (Fig. 4). Second contribution to κ_l^{xx} is entirely due to optical phonons and it is slightly smaller than the first one, but still constitutes a 30-45% of the total value. While acoustic phonons with long mean free paths usually control κ_l , this is one of the few reported examples in which high-frequency optical modes strongly modify κ_l .⁴⁴ Even more interestingly, optical modes contributions to κ_l^{xx} are different for each material. BiOCuS presents the lowest κ_l^{xx} because the optical mode contribution to κ_l is proportionally smaller than in

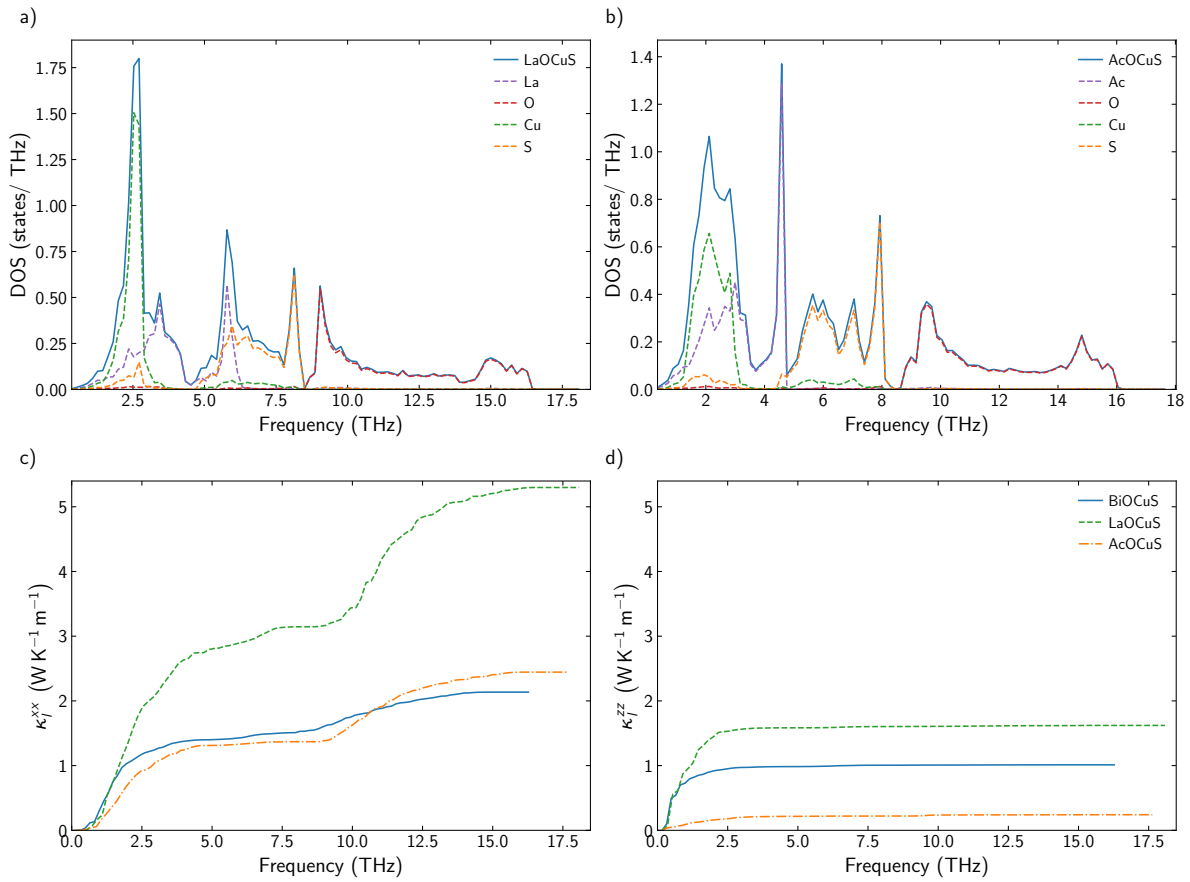


Fig. 3 (a-b) Total and projected phonon density of states, DOS, of LaOCuS and AcOCuS. (c-d) Cumulative κ_l^{xx} and κ_l^{zz} for BiOCuS, LaOCuS, and AcOCuS at 300 K.

LaOCuS and AcOCuS. Due to these high frequency contributions, κ_l^{xx} is not ranked by the atomic mass of the trivalent atom. This behavior is directly connected to the scattering rates of the vibrational modes at high frequencies (Fig. 4). LaOCuS presents the lower scattering rates for frequencies larger than 7-8 THz, which is consistent with larger contributions to κ_l . On the other hand, BiOCuS exhibits the highest scattering rates at high frequencies, so their contributions to κ_l is reduced. Projected scattering rates show how this high-frequency contribution to κ_l is mainly localized on O atoms vibrations. Thus, the differences in the scattering rates of the high frequency phonons stem from the different nature of the M-O bonds (M=La, Ac, Bi). If BiOCuS is the material with the highest scattering rates at high frequencies, atomic mass cannot be the only variable to consider and its lone pair electron must play a role. It has been proven that the presence of stereochemically active lone pair electrons, LPEs, enhances the anharmonicity of a system and reduce κ_l .⁴⁵ This reduction is due to nonlinear repulsive electrostatic force between LPEs and other neighboring bonds, which lowers the lattice symmetry and promote the appearance of more scattering processes.^{46,47} Following the revised lone pair model,⁴⁸ it has been experimentally confirmed the presence of a LPE in BiOCuSe.⁴⁹ This LPE is the result of the formation of a Bi 6s-O 2p bonding state just below the valence band and the stabilization of the Bi 6s-O 2p anti-bonding state due to its

hybridization with Bi 6p empty states. This hybridization has also been described through first principles calculations.^{50,51} Here, it is important to stress the relevance of the concept of LPE as a bonding state, in order to understand their role reducing κ_l . Although, optical phonons contributions to κ_l^{xx} are mainly projected on O atoms, the scattering rates of these vibrational modes are larger than in LaOCuS and AcOCuS because of the Bi-O bond nature, which is singularly characterized by the presence of a LPE.

The role of Cu. Some previous works have described Cu as the the main contributor to the low lattice thermal conductivity of BiOCuQ family.²⁵ Vaquero *et al.* used neutro diffraction experiments and DFT calculations to demonstrate that low-energy vibrational modes on Cu and its weak bonding with the chalcogen are essential to understand their low κ_l . However, is this a singular interaction between Cu and the chalcogens or is it possible to obtain similar results with other elements? Here, the vibrational properties and κ_l of LaOMS (M = Cu, Ag, Li) have been explored for a better understanding of the role of the monovalent cation. While the substitution of Bi by Ac or La do not produce large modification in the phonon DOS of the compounds, substituting Cu by Li or Ag lead to severe modifications of the vibrational structure (Fig. 5a-b). Li atoms contributions to the DOS are shifted more than 7 THz, if compared with the projection of Cu atoms in LaOCuS DOS. On the other hand, Ag substitution shift the mono-

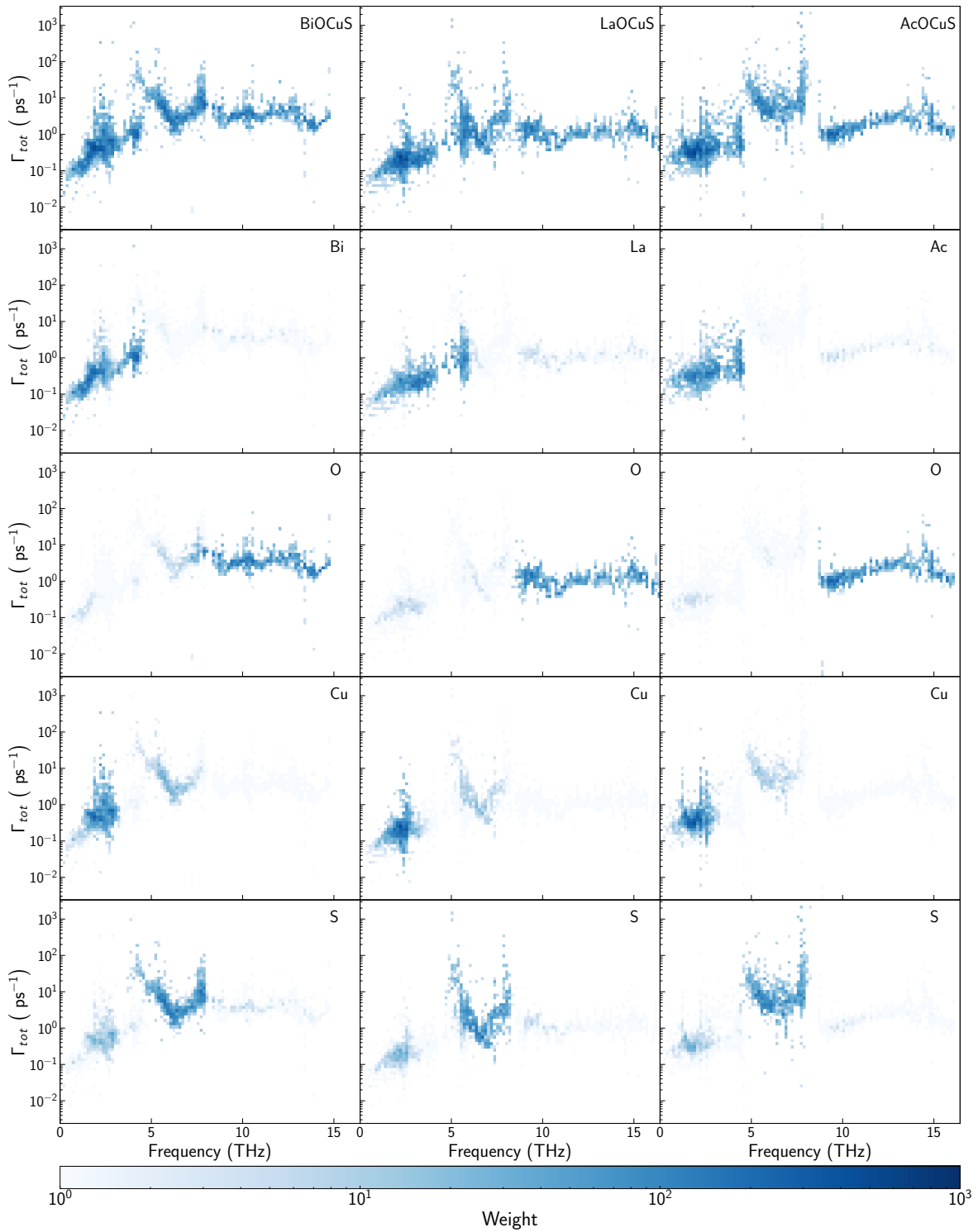


Fig. 4 Scattering rates - frequency heat maps for BiOCuSe (top left), LaOCuS (top mid), and AcOCuS (top right) at 300 K. Heat map color represents the density of vibrational modes with similar scattering rates and frequencies. Projections on each element have been depicted below the total contributions.

valent contribution to the DOS to even lower frequencies, making it as the main contributor to the acoustic modes. The absence of a heavy monovalent atom such as Cu or Ag also increases the group velocities of the phonons at low frequencies for LaOLiS (Fig. 5c). This behaviour has a clear impact in the cumulative κ

of LaOLiS which grows rapidly up to 6 W K⁻¹ m⁻¹ below 4 THz (Fig. 5d). Silver substitution produces the opposite effect on κ because acoustic-optical scattering region is pushed to even lower frequencies when compared to LaOCuS. That trend can be visualized in the scattering rates heat maps where LaOAgS scattering

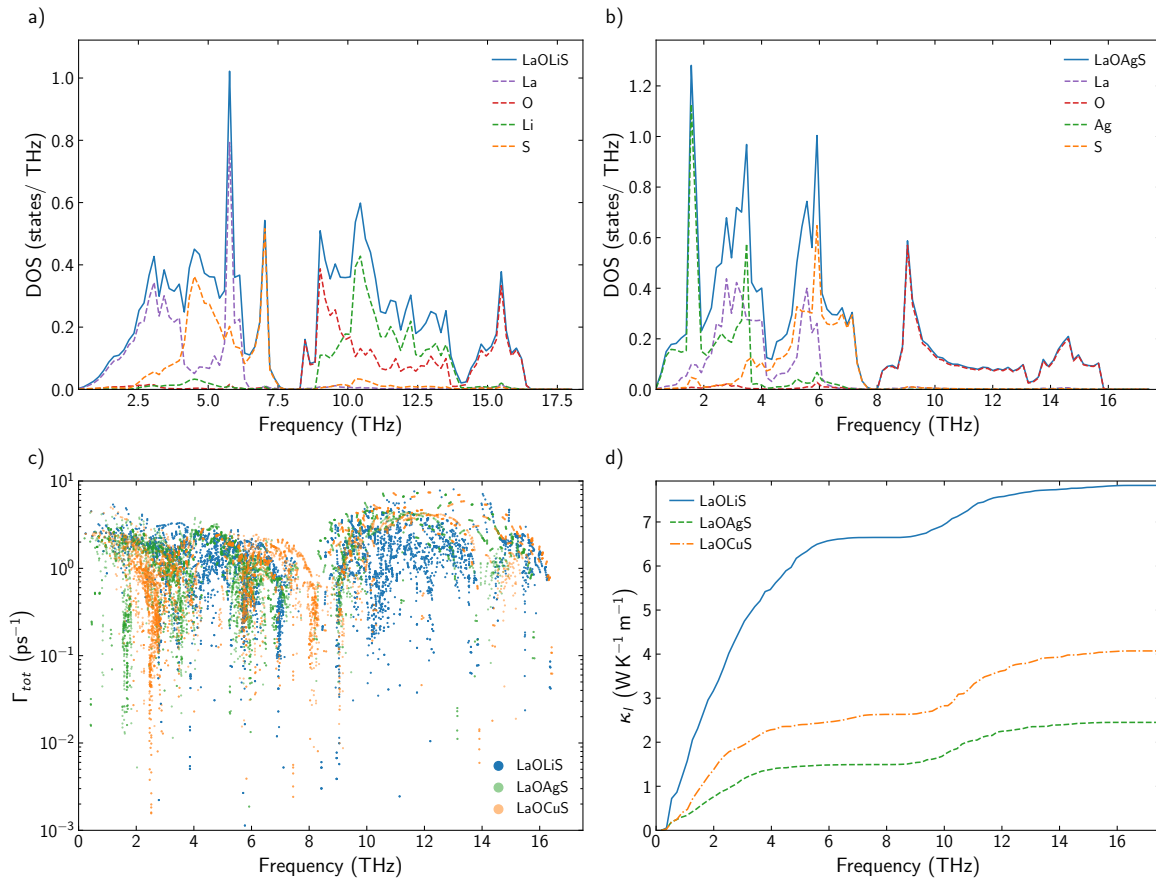


Fig. 5 (a-b) Total and projected phonon density of states, DOS, of LaOLiS and LaOAgS. (c) Group velocities, v_g for LaOLiS (blue), LaOAgS (green), and LaOCuS (orange). (d) Cumulative κ_l for LaOLiS (blue solid line), LaOAgS (green dashed line) and LaOCuS (orange dashed line).

rates at low frequencies are higher than in LaOCuS and LaOLiS. As expected, La and Ag contributions dominates in this region so the anharmonicity comes from interlayer scattering (Fig. 6).

Designing new oxychalcogenides. Understanding the mechanisms that governs κ_l in oxychalcogenides become fundamental for the design of new materials with the potential TE applications. Larger contribution to κ_l corresponds to scattering processes between acoustic and low frequency optical phonons. This scattering can be maximized with heavy monovalent and trivalent atoms and using Se or Te. In addition, second contribution to κ_l at higher frequencies can be minimized using trivalent atoms with LPE that increases the scattering rates of high frequency phonons, mainly localized on O atoms. Taking both considerations into account, BiOAgSe is proposed as an oxychalcogenide with potential ultra-low κ_l . To the best of our knowledge, there are very few recent reports in which BiOAgSe has been synthesized.^{35,52,53} Some of these works highlight their optoelectronic properties and its potential photoconversion applications.^{35,52} Although BiOAgTe should present also a very low κ ,^{54,55} selenides are usually preferred instead of tellurides because Tellurium is a very volatile element which hampers the stability and durability of the material. BiOAgSe has been very recently reported as a promising candidate as TE materials due to its tunable electrical conductivity,⁵⁶ and, during the preparation of this manuscript, its κ_l has been

reported using the relaxation time approximation, RTA.^{53,57} However, very little has been described about the origin of its ultra-low κ_l . Phonon DOS already shows how there are some important differences with respect previous materials (Fig. 7a). BiOAgSe exhibits κ_l lower than 1.2 W K⁻¹ m⁻¹ at 300 K in both xx and zz directions and an average κ_l of 0.87 W K⁻¹ m⁻¹ (Fig. 7b-c). These values are slightly larger than the ones predicted by Li *et al.*⁵⁷ because the RTA tends to underestimate κ_l . This low values are due to the effective overlap between acoustic and optical modes mainly projected on the Bi and Ag atoms (Fig. 7a and Fig. 6). Second contribution to κ_l at high frequencies is also very small (around 0.3 W K⁻¹ m⁻¹) (Fig. 7c) and it is minimized with respect to other oxychalcogenides explored in this work due to the high scattering rates of the high frequency phonons (Fig. 6). In addition to the single-crystal lattice thermal conductivity calculated by Li *et al.* and the analysis of its origin presented in this work, lattice thermal conductivity for polycrystalline BiOAgSe is included (Fig. 7d). The effect of nanostructuring on the thermal conductivity is based on the decomposition of the contributions to κ by the phonon mean free path.⁵⁸ In this approach, the value of κ_l corresponding to a particular particle size L is calculated as the cumulative contributions for all mean free paths up to L . Large scattering rates or very short lifetimes produces short mean-free-path so small grains are needed to major reductions of κ_l . However, some reduction can

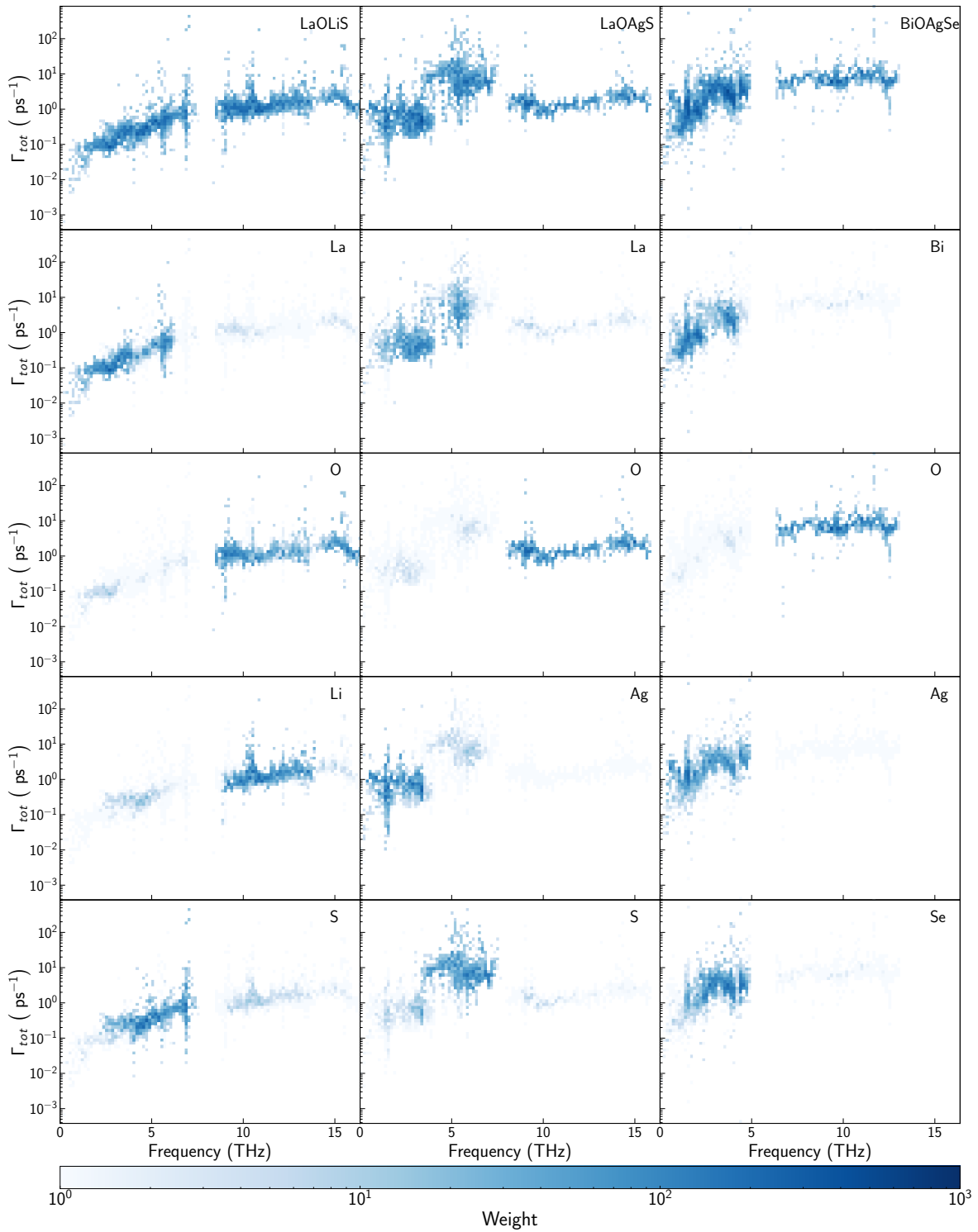


Fig. 6 Scattering rates - frequency heat maps for LaOLiS (top left), LaOAgS (top mid), and BiOAgSe (top right). Heat map color represents the density of vibrational modes with similar scattering rates and frequencies. Projections on each element have been depicted below the total contributions.

be achieved if the average grain size is reduce to 100 nm, where the value of κ_l is below $0.8 \text{ W K}^{-1} \text{ m}^{-1}$.

4 Conclusions

In this work, we have demonstrated how machine learning not only accelerates the prediction of the lattice thermal conductivity

for large chemical spaces with high accuracy but also catalyzes the development of design principles to discover new thermoelectric materials. Lattice thermal conductivity of seven oxychalcogenides with very different compositions have been predicted and analyzed. While most previous theoretical work discussed the origin of the low κ_l of oxychalcogenides based on harmonic properties, we

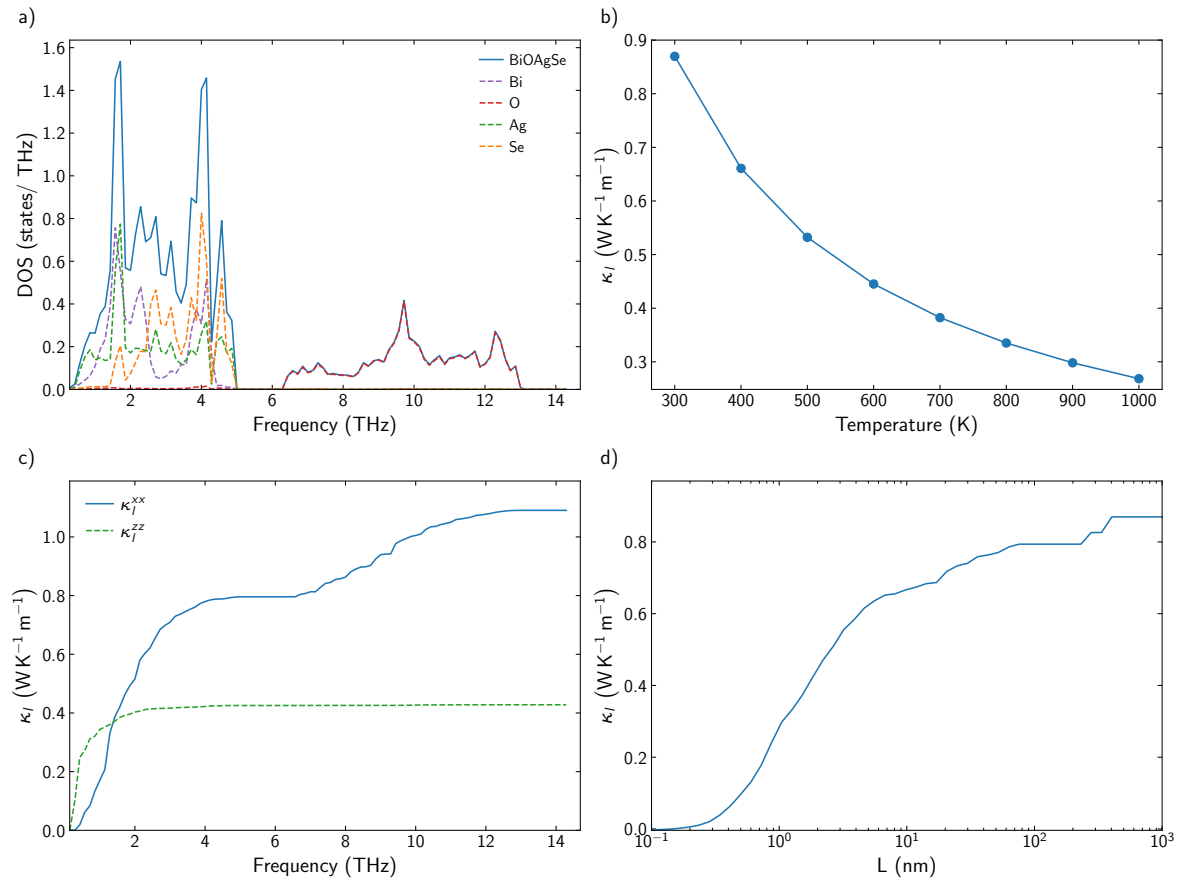


Fig. 7 (a) Total and projected phonon density of states, DOS, of BiOAgSe. (b) Average κ_l of BiOAgSe between 300 K and 1000 K. (c) Cumulative κ_l^{xx} and κ_l^{zz} for BiOAgSe at 300 K. (d) Cumulative average lattice thermal conductivity at 300 K from mean-free path contributions up to distance L for BiOAgSe, indicating the effect that nanostructuring would have on their κ_l .

introduce a new approach based on the analysis of phonon DOS, cumulative κ_l , and most importantly scattering rates heat maps and their atomic projections. Chemical substitution for all sites have been considered in order to unravel the role of each chemical specie. Two main contributions to κ_l have been described. Low frequency phonons contribution to κ_l can be minimized with heavy trivalent and monovalent cations. Heavy atoms with similar vibrational frequencies promotes the scattering between acoustic and low-frequency optical modes. We also found that high-frequency optical modes have an important contribution to κ_l^{xx} . Atomic projections of the scattering rates heat maps show that these modes are mainly localized on the O atoms. Although O atoms are the main contributors to these high-frequency phonons, their scattering rates are sensitive to the trivalent atom that is part of the $[M_2O_2]^{2+}$ slab. It has been demonstrated that the atomic mass of the trivalent atom is not the driven force that modifies the scattering rates of these modes. However, it is the element with the LPE, the one that increases the scattering rates and reduces the contribution of these modes to κ_l . Finally, we have followed these design principles based on chemical composition to propose BiOAgSe as one of the best oxychalcogenides for TE applications.

Author Contributions

The authors confirm contribution to the paper as follows: Conceptualization: J.J.P. and A.M.M.; Data curation: E.J.B and J.S.; Investigation: E.J.B, J.S. and F.L.P.; Visualisation: E.J.B.; Writing – original draft: J.J.P.R., A.M.M. and J.F.S. All authors reviewed the results and approved the final version of the manuscript.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

This work was funded by the Ministerio de Ciencia e Innovación (PID2019-106871GB-I00). E.J.B., J.J.P., J.S., A.M.M., and J.F.S. thankfully acknowledge the computer resources at Lusitania and the technical support provided by Cénits COMPUTAEX and Red Española de Supercomputación, RES (QS-2021-1-0027).

Notes and references

- 1 H.-S. Kim, Z. Gibbs, Y. Tang, H. Wang and G. Snyder, *APL Mater.*, 2015, **3**, 041506.
- 2 L. Xi, S. Pan, X. Li, Y. Xu, J. Ni, X. Sun, J. Yang, J. Luo, J. Xi, W. Zhu, X. Li, D. Jiang, R. Dronskowski, X. Shi, G. Snyder and W. Zhang, *J. Am. Chem. Soc.*, 2018, **140**, 10785–10793.

- 3 T. Zhu, Y. Liu, C. Fu, J. Heremans, J. Snyder and X. Zhao, *Adv. Mater.*, 2017, **29**, 1605884.
- 4 P. Gorai, V. Stevanović and E. S. Toberer, *Nat. Rev. Mater.*, 2017, **2**, 17053.
- 5 C. Carbogno, R. Ramprasad and M. Scheffler, *Phys. Rev. Lett.*, 2017, **118**, 175901.
- 6 J. J. Plata, P. Nath, D. Usanmaz, J. Carrete, C. Toher, M. de Jong, M. D. Asta, M. Fornari, M. Buongiorno Nardelli and S. Curtarolo, *NPJ Comput. Mater.*, 2017, **3**, 45.
- 7 C. Verdi, F. Karsai, P. Liu, R. Jinnouchi and G. Kresse, *NPJ Comput. Mater.*, 2021, **7**, 156.
- 8 F. Eriksson, E. Fransson and P. Erhart, *Adv. Theory Simul.*, 2019, **2**, 1800184.
- 9 E. Fransson, F. Eriksson and P. Erhart, *npj Comput. Mater.*, 2020, **6**, 135.
- 10 Y. Iwasaki, I. Takeuchi, V. Stanev, A. Kusne, M. Ishida, A. Kirihara, K. Ihara, R. Sawada, K. Terashima, H. Someya, K.-I. Uchida, E. Saitoh and S. Yorozu, *Sci. Rep.*, 2019, **9**, 2751.
- 11 H. Zhu, C. Xiao and Y. Xie, *Adv. Mater.*, 2018, **30**, 1802000.
- 12 S. Luu and P. Vaqueiro, *J. Materiomics*, 2016, **2**, 131–140.
- 13 Y. Liu, L.-D. Zhao, Y. Liu, J. Lan, W. Xu, F. Li, B.-P. Zhang, D. Berardan, N. Dragoe, Y.-H. Lin, C.-W. Nan, J.-F. Li and H. Zhu, *J. Am. Chem. Soc.*, 2011, **133**, 20112–20115.
- 14 F. Li, J.-F. Li, L.-D. Zhao, K. Xiang, Y. Liu, B.-P. Zhang, Y.-H. Lin, C.-W. Nan and H.-M. Zhu, *Energy Environ. Sci.*, 2012, **5**, 7188–7195.
- 15 Y.-L. Pei, J. He, J.-F. Li, Fuli, Q. Liu, W. Pan, C. Barreteau, D. Berardan, N. Dragoe and L.-D. Zhao, *NPG Asia Mater.*, 2013, **5**, e47.
- 16 M.-K. Han, Y.-S. Jin, B. Yu, W. Choi, T.-S. You and S.-J. Kim, *J. Mater. Chem. A*, 2016, **4**, 13859–13865.
- 17 H. Zhu, T. Su, H. Li, C. Pu, D. Zhou, P. Zhu and X. Wang, *J. Eur. Ceram. Soc.*, 2017, **37**, 1541–1546.
- 18 T.-L. Chou, G. Tewari, D. Srivastava, A. Yamamoto, T.-S. Chan, Y.-Y. Hsu, J.-M. Chen, H. Yamauchi and M. Karppinen, *Mater. Chem. Phys.*, 2016, **177**, 73–78.
- 19 C. Barreteau, D. Bérardan, L. Zhao and N. Dragoe, *J. Mater. Chem. A*, 2013, **1**, 2921–2926.
- 20 G. Li, S. Hao, S. Morozov, P. Zhai, Q. Zhang, W. Goddard and G. Snyder, *ACS Appl. Mater. Interfaces*, 2018, **10**, 6772–6777.
- 21 M. Toriyama, J. Qu, G. Snyder and P. Gorai, *J. Mater. Chem. A*, 2021, **9**, 20685–20694.
- 22 J. Ding, B. Xu, Y. Lin, C. Nan and W. Liu, *New J. Phys.*, 2015, **17**, 083012.
- 23 S. Saha, *Phys. Rev. B*, 2015, **92**, 041202.
- 24 S. Saha and G. Dutta, *Phys. Rev. B*, 2016, **94**, 125209.
- 25 P. Vaqueiro, R. Al Orabi, S. Luu, G. Guérou, A. Powell, R. Smith, J.-P. Song, D. Wee and M. Fornari, *Phys. Chem. Chem. Phys.*, 2015, **17**, 31735–31740.
- 26 G. Kresse and J. Hafner, *Phys. Rev. B*, 1993, **47**, 558–561.
- 27 G. Kresse and J. Furthmüller, *Phys. Rev. B*, 1996, **54**, 11169–11186.
- 28 P. E. Blöchl, *Phys. Rev. B*, 1994, **50**, 17953–17979.
- 29 J. P. Perdew, K. Burke and M. Ernzerhof, *Phys. Rev. Lett.*, 1996, **77**, 3865–3868.
- 30 S. Grimme, J. Antony, S. Ehrlich and H. Krieg, *J. Chem. Phys.*, 2010, **132**, 154104.
- 31 C. E. Calderon, J. J. Plata, C. Toher, C. Oses, O. Levy, M. Fornari, A. Natan, M. J. Mehl, G. L. W. Hart, M. Buongiorno Nardelli and S. Curtarolo, *Comput. Mater. Sci.*, 2015, **108 Part A**, 233–238.
- 32 A. Kusainova, P. Berdonosov, L. Akselrud, L. Kholodkovskaya, V. Dolgikh and B. Popovkin, *J. Solid State Chem.*, 1994, **112**, 189–191.
- 33 Y. Takano, C. Ogawa, Y. Miyahara, H. Ozaki and K. Sekizawa, *J. Alloys Compd.*, 1997, **249**, 221–223.
- 34 M. Palazzi, C. C. P. Laruelle and J. Flahaut, *The Rare Earths in Modern Science and Technology*, Boston, MA, 1982, pp. 347–350.
- 35 J. Gamon, D. Giaume, G. Wallez, J.-B. Labégorre, O. Lebedev, R. Al Rahal Al Orabi, S. Haller, T. Le Mercier, E. Guilmeau, A. Maignan and P. Barboux, *Chem. Mater.*, 2018, **30**, 549–558.
- 36 J. J. Plata, V. Posligua, A. M. Marquez, J. F. Sanz and R. Grau-Crespo, *Chem. Mater.*, 2022, **34**, xxxxxxxx.
- 37 J. J. Plata, V. Posligua, A. M. Marquez, J. Fdez Sanz and R. Grau-Crespo, *hiPhive Wrapper*, 2021, https://github.com/NewMaterialsLab/hiPhive_wrapper_NML.
- 38 W. Li, J. Carrete, N. A. Katcho and N. Mingo, *Comput. Phys. Commun.*, 2014, **185**, 1747–1758.
- 39 A. Togo and I. Tanaka, *Scr. Mater.*, 2015, **108**, 1–5.
- 40 E. J. Blancas, J. J. Plata, A. M. Marquez and J. Fdez Sanz, *Scattering rates heat map plotter*, 2022, <https://github.com/ErnstBlancas/PRmaps>.
- 41 S. Kumar and U. Schwingenschlögl, *Phys. Chem. Chem. Phys.*, 2016, **18**, 19158–19164.
- 42 H. S. Ji, A. Togo, M. Kaviany, I. Tanaka and J. H. Shim, *Phys. Rev. B*, 2016, **94**, 115203.
- 43 M. Nielsen, V. Ozolins and J. Heremans, *Energy Environ. Sci.*, 2013, **6**, 570–578.
- 44 Z. Li, H. Xie, S. Hao, Y. Xia, X. Su, M. G. Kanatzidis, C. Wolverton and X. Tang, *Phys. Rev. B*, 2021, **104**, 245209.
- 45 C. Chang and L.-D. Zhao, *Mater. Today Phys.*, 2018, **4**, 50–57.
- 46 D. Morelli, V. Jovovic and J. Heremans, *Phys. Rev. Lett.*, 2008, **101**, 035901.
- 47 E. Skoug and D. Morelli, *Phys. Rev. Lett.*, 2011, **107**, 235901.
- 48 A. Walsh, D. Payne, R. Egdell and G. Watson, *Chem. Soc. Rev.*, 2011, **40**, 4455–4463.
- 49 S. Sallis, L. Piper, J. Francis, J. Tate, H. Hiramatsu, T. Kamiya and H. Hosono, *Phys. Rev. B*, 2012, **85**, 085207.
- 50 E. Stampler, W. Sheets, M. Bertoni, W. Prellier, T. Mason and K. Poeppelmeier, *Inorg. Chem.*, 2008, **47**, 10009–10016.
- 51 H. Hiramatsu, H. Yanagi, T. Kamiya, K. Ueda, M. Hirano and H. Hosono, *Chem. Mater.*, 2008, **20**, 326–334.
- 52 A. Baqais, A. Curutchet, A. Ziani, H. Ait Ahsaine, P. Sautet, K. Takanabe and T. Le Bahers, *Chem. Mater.*, 2017, **29**, 8679–8689.
- 53 C. Zhang, J. He, R. McClain, H. Xie, S. Cai, L. Walters, J. Shen,

- F. Ding, X. Zhou, C. Malliakas, J. Rondinelli, M. Kanatzidis, C. Wolverton, V. Dravid and K. Poeppelmeier, *J. Am. Chem. Soc.*, 2022, **144**, 2569–2579.
- 54 M. Mukherjee and A. Singh, *ACS Applied Materials and Interfaces*, 2020, **12**, 8280–8287.
- 55 J. He, Z. Yao, V. Hegde, S. Naghavi, J. Shen, K. Bushick and C. Wolverton, *Chem. Mater.*, 2020, **32**, 8229–8242.
- 56 J. Li, C. Zhang, Y. Yan, J. Yang, B. Shi, Y. Wang and Z. Cheng, *Comp. Mater. Sci.*, 2020, **171**, 109273.
- 57 J. Li, W. Zhai, C. Zhang, Y. Yan, P.-F. Liu and G. Yang, *Mater. Adv.*, 2021, **2**, 4876–4882.
- 58 C. Dames and G. Chen, *Thermoelectrics Handbook: Macro to Nano*, Boca Raton, FL, 2006, p. 421.